

LAB 6

DYNAMIC ROUTING CONFIGURATION: RIP, EIGRP, OSPF, BGP

OBJECTIVES

- To understand the working principles of RIP, EIGRP, OSPF, and BGP routing protocols.
- To configure dynamic routing protocols on routers using Cisco Packet Tracer.
- To enable automatic route discovery between different networks.
- To establish communication between networks using dynamic routing configurations.

THEORY

1. RIP (Routing Information Protocol)

Operation: Uses distance-vector algorithm with hop count as the metric (maximum 15 hops). Sends complete routing table updates every 30 seconds to neighboring routers. Uses Bellman-Ford algorithm for route calculation.

Application: Best suited for small, simple networks with limited size. Commonly used in small office/home office (SOHO) environments and educational labs due to simple configuration.

Limitation: Limited to 15 hops maximum (16 is unreachable), slow convergence time, inefficient bandwidth usage due to periodic full updates, no support for VLSM in RIPv1, susceptible to routing loops.

2. EIGRP (Enhanced Interior Gateway Routing Protocol)

Operation: Advanced distance-vector protocol using DUAL (Diffusing Update Algorithm). Calculates metric based on bandwidth, delay, load, and reliability. Sends partial, triggered updates only when topology changes. Maintains neighbor relationships using hello packets.

Application: Ideal for large enterprise networks, multi-vendor environments (with redistribution), networks requiring fast convergence and scalability. Supports hierarchical network designs.

Limitation: Cisco proprietary (though partially opened), complex configuration compared to RIP, requires more CPU and memory resources, can be difficult to troubleshoot in large networks.

3. OSPF (Open Shortest Path First)

Operation: Link-state protocol that builds complete network topology database. Uses Dijkstra's SPF algorithm to calculate shortest path. Organizes networks into areas with Area 0 as backbone. Sends link-state advertisements (LSAs) and maintains neighbor adjacencies.

Application: Best for large, complex enterprise networks requiring scalability and fast convergence. Supports hierarchical design with areas, efficient for networks with redundant paths. Industry standard for IGP.

Limitation: Complex configuration and troubleshooting, high CPU and memory consumption during SPF calculations, requires careful network design with proper area planning, steep learning curve for administrators.

4. BGP (Border Gateway Protocol)

Operation: Path-vector exterior gateway protocol that exchanges routing information between autonomous systems (AS). Uses TCP port 179 for reliable connections. Makes routing decisions based on policies, network paths, and rules rather than just metrics. Maintains peer relationships.

Application: Essential for internet routing and connecting organizations to ISPs. Used for multi-homed connections, traffic engineering, policy-based routing between different organizations or autonomous systems.

Limitation: Very complex configuration requiring deep understanding, slow convergence compared to IGPs, requires significant memory for full internet routing table, prone to misconfigurations causing internet-wide issues, not suitable for internal enterprise routing.

NETWORK TOPOLOGY

A network topology was created using two routers: Router2 (1841) and Router1 Suraj (1841), interconnected via a serial/WAN link using their serial interfaces. The topology consists of two separate Local Area Networks (LANs) connected through these routers.

Switch0 (2960-24TT) is connected to Router2 and provides connectivity to two end devices: PC0 (PC-PT) and PC1 (PC-PT), forming Network 192.168.1.0/24. Switch1 (2960-24TT) is connected to Router1 Suraj and provides connectivity to two end devices: PC2 (PC-PT) and PC3 (PC-PT), forming Network 192.168.2.0/24.

The serial connection between Router2 and Router1 Suraj enables inter-network communication, allowing devices in the 192.168.1.0/24 network to communicate with devices in the 192.168.2.0/24 network through dynamic routing protocols. Green indicators on the connections represent active and functioning network links in the Cisco Packet Tracer simulation environment.

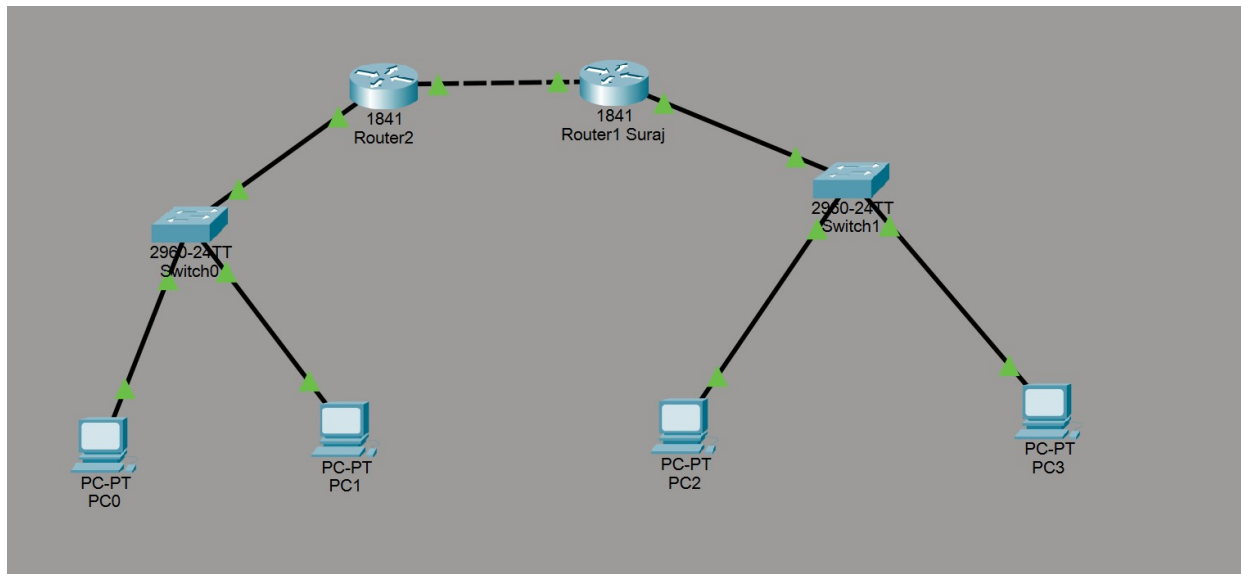


Fig: Network Topology

Configuration Table:

Network devices were configured with IP addresses and subnet masks as detailed in the table below:

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router2	Gig0/1	192.168.1.1	255.255.255.0	N/A
Router2	Gig0/0	10.0.0.1	255.0.0.0	N/A
Router1 Suraj	Gig0/1	192.168.2.1	255.255.255.0	N/A
Router1 Suraj	Gig0/0	10.0.0.2	255.0.0.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
PC2	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.3	255.255.255.0	192.168.2.1

1. RIP Configuration Commands

Router2 (Left sided one in topology):

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

Note: If you just see “Router >” instead of “Router(config)#” (as in the first line), then execute the following commands also.

```
Router>enable
Router# configure terminal
Router(config)#
```

Router1 (Right sided one in topology):

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
Router(config-router)# no auto-summary
```

2. EIGRP Configuration Commands

Router2 (Left sided one in topology):

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.1.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

Router1 (Right sided one in topology):

```
Router(config)# router eigrp 100
Router(config-router)# network 192.168.2.0 0.0.0.255
Router(config-router)# network 10.0.0.0 0.0.0.255
Router(config-router)# no auto-summary
```

3. OSPF Configuration Commands

Router2 (Left sided one in topology):

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.1.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

Router1 (Right sided one in topology):

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.2.0 0.0.0.255 area 0
Router(config-router)# network 10.0.0.0 0.0.0.255 area 0
```

4. BGP Configuration Commands

Router2 (Left sided one in topology) (AS 65001):

```
Router(config)# router bgp 65001
Router(config-router)# neighbor 10.0.0.2 remote-as 65002
Router(config-router)# network 192.168.1.0 mask 255.255.255.0
```

Router1 (Right sided one in topology) (AS 65002):

```
Router(config)# router bgp 65002
Router(config-router)# neighbor 10.0.0.1 remote-as 65001
Router(config-router)# network 192.168.2.0 mask 255.255.255.0
```

RESULT

The dynamic routing protocols were successfully implemented and tested across the network topology. Connectivity verification was performed by sending ping requests from PC2 (192.168.2.2) in the 192.168.2.0/24 network to PC0 (192.168.1.2) in the 192.168.1.0/24 network. Each routing protocol (RIP, EIGRP, OSPF, and BGP) demonstrated successful automatic route propagation and routing table updates. All packets were transmitted and received without loss, confirming proper inter-network communication and validating the correct configuration of the dynamic routing protocols.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping (RIP configuration)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: Ping (EIGRP)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping (OSPF)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time<1ms TTL=126
Reply from 192.168.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Fig: Ping (BGP)

DISCUSSION

The laboratory experiment successfully demonstrated the implementation of four dynamic routing protocols using Cisco Packet Tracer. RIP provided the simplest configuration but exhibited slower convergence due to periodic updates. EIGRP demonstrated faster convergence through triggered updates and efficient bandwidth utilization. OSPF required more complex configuration with area designation but offered vendor-neutral standardization. BGP presented the highest configuration complexity with autonomous system requirements. Connectivity verification through ping tests between PC2 and PC0 confirmed successful automatic route discovery and inter-network communication across all protocols.

CONCLUSION

This laboratory exercise successfully accomplished the objectives of understanding and configuring dynamic routing protocols on Cisco routers. All four protocols (RIP, EIGRP, OSPF, and BGP) were properly implemented and verified through successful connectivity tests. The experiment demonstrated automatic route discovery and routing table maintenance without manual intervention.